

End-to-end DS pipeline documentation

Task – 4 : Week 3

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Technical Methodology Note: Integrated Population Health Analytics

1. Pipeline Overview :

This report presents the findings of a comprehensive end-to-end data pipeline designed to enhance service delivery and financial stewardship. By integrating Talking Therapies (IAPT) demand metrics with national prescribing data and deprivation indices, we have moved from retrospective reporting to a proactive, data-driven management model.

Workstream	Model Type	Core Objective
Demand Forecast	ARIMA (1,1,0)	Predictive capacity planning for IAPT referrals.
Anomaly Detection	Isolation Forest	Identification of CCG prescribing outliers.
Spatial Analysis	Bivariate Mapping	Identifying "High-Need Low-Service" inequalities.

2. Detailed Model Documentation :

2.1 IAPT Demand Forecasting :

- Input: Monthly Time Series (Monthly_Time_Series_March25_March26.csv).

- Assumptions: Assumes historical seasonal patterns (e.g., winter surges) will continue.
- Limitations: Does not account for sudden "Black Swan" events (e.g., new pandemics or major policy shifts).
- Output: Point estimates for next 3 months with 95% Confidence Intervals.

2.2 Prescribing Anomaly Detection :

- Input: CCG-level aggregation of ITEMS and ACTUAL_COST.
- Methodology: Isolation Forest (Unsupervised Learning).
- Assumptions: Presumes that the majority of prescribing behavior is "normal" and that outliers are mathematically distinct.
- Limitations: High spend does not always equal poor practice; clinical context (specialist clinics) is required for final validation.

2.3 Geo-Spatial Health Equity :

- Input: Sub-ICB Access rates joined with ONS Index of Multiple Deprivation (IMD).
- Methodology: Spatial Join & Bivariate Classification.
- Output: Priority list of under-served, high-deprivation areas.

3. Data Governance & Processing :

To ensure this pipeline meets NHS Information Governance (IG) standards:

- Aggregation: All 6.8 GB raw files were processed using "Chunking" to prevent memory overflow and aggregated to CCG level to ensure no Patient Identifiable Information (PII) was processed.
- Reproducibility: All models are built using standard Python libraries (pandas, statsmodels, scikit-learn) to ensure they can be audited by internal NHS analysts.
- Validation: Anomaly flags were cross-referenced against 7 months of data to reduce false-positive rates.

4. Assumptions :

Here are the detailed assumptions for the three workstreams:

1. Demand Forecasting (ARIMA Assumptions) :

- **Linearity of Time:** The model assumes that future referral counts are a linear function of past observations and past errors. It does not account for exponential growth or non-linear surges.
- **Stationarity (after Differencing):** We assume that by "differencing" the data (looking at the change between months rather than raw totals), we have removed the trend and seasonality. The model assumes the underlying "process" generating the referrals remains stable over time.
- **Continuity of External Drivers:** Since this is a univariate model (using only referral data), it assumes that external factors—such as the number of GPs in a region or the prevalence of mental health conditions—remain constant or continue to change at the same historical rate.
- **Error Normality:** The model assumes that the "residuals" (the difference between the forecast and actuals) are normally distributed ("White Noise"). If the errors show a pattern, the model is missing a key piece of information.

2. Anomaly Detection (Isolation Forest Assumptions)

- **The "Few and Different" Rule:** The fundamental assumption of an Isolation Forest is that anomalies are **quantitatively few** (low frequency) and **qualitatively different** (distant in feature space). If "bad" prescribing becomes the majority behavior, the model will incorrectly flag "good" prescribing as the anomaly.
- **Feature Independence:** The model assumes that ITEMS and ACTUAL_COST provide enough information to define "normal" behavior. It assumes no other hidden variables (like the specific BNF drug chapter) are required to identify a statistical outlier at the first gate.
- **Uniformity of CCG Structure:** The model assumes that CCGs are comparable units. It assumes that a "normal" relationship between cost and volume in a rural CCG should look relatively similar to that in an urban CCG, unless the data proves otherwise.

3. Geo-Spatial Analysis (Spatial Join Assumptions)

- **Homogeneity of Deprivation:** The "Spatial Join" assumes that the average IMD score of a Sub-ICB is a fair representation of the "Need" for that entire area. It assumes that internal variances within the geography (richer and poorer streets) do not cancel each other out in a way that hides the true clinical demand.
- **Access as a Proxy for Service Quality:** This analysis assumes that "Count of Patients Accessing Services" is the primary indicator of service success. It assumes that a high volume of access is inherently "good," without accounting for the quality or outcome of the treatment provided.
- **Lookup Accuracy:** It assumes that the ONS LSOA-to-Sub-ICB lookup table is current and that the boundaries used in the IAPT reporting align perfectly with the boundaries used in the Deprivation Index reporting.

4. General Pipeline Assumptions

- **Data Integrity:** We assume that the monthly 6.8 GB files have been cleaned at the source. We assume that a "zero" in the data represents a true zero and not a "missing value" or a technical failure in the reporting practice.
- **Systemic Stability:** We assume that the definition of a "Referral" or a "Prescribed Item" has not changed between January 2025 and July 2025. If the NHS changed the way these are counted mid-year, the model would perceive it as a trend or an anomaly rather than a clerical change.

5. Limitation's:

Here are the detailed limitations for each component of the pipeline:

1. IAPT Demand Forecast (ARIMA Model)

- **Stationarity Assumption:** The ARIMA model assumes that future patterns will mirror historical ones. It cannot predict "Structural Breaks"—such as a sudden change in NHS policy, a new global health crisis, or the sudden closure of a large local provider.
- **Lack of External Covariates:** This is a univariate model, meaning it only looks at past referral numbers. It does not "know" about external factors like local unemployment rates, changes in cost-of-living, or winter flu spikes, all of which correlate with mental health demand.
- **Confidence Interval Width:** As we project further into the future (e.g., June 2026), the "uncertainty" grows. While the point estimate is 161k, the model

acknowledges that the reality could be anywhere between 112k and 210k. Relying solely on the middle number for staffing is a risk.

2. Prescribing Anomaly Detection (Isolation Forest)

- **The "Context Gap":** The model is mathematically focused, not clinically focused. It flags a CCG as an "anomaly" if its spending is statistically distant from the mean. However, an area might have high costs because it hosts a specialized tertiary center (e.g., an oncology hub) that *rightfully* prescribes expensive medications. These are "True Anomalies" but not "Actionable Faults."
- **Contamination Parameter Bias:** We set the contamination to 0.01 (1%). This is an arbitrary threshold. If 5% of CCGs actually have poor prescribing habits, the model will miss 4% of them. Conversely, if all CCGs are performing perfectly, the model will still force itself to flag the "weirdest" 1%, creating false positives.
- **Sensitivity to Data Quality:** In large datasets (like the 6.8 GB prescribing files), "Anomalies" are often just data entry errors (e.g., a misplaced decimal point in a cost field) rather than actual prescribing behavior.

3. Geo-Spatial Analysis (IMD Spatial Join)

- **The Ecological Fallacy:** This analysis assumes that because a Sub-ICB area is "deprived" on average, everyone living there has a high need. In reality, wealthy pockets exist within deprived CCGs and vice-versa. Aggregating to the Sub-ICB level can mask these hyper-local health inequalities.
- **Modifiable Areal Unit Problem (MAUP):** The results might look completely different if we mapped them by "Primary Care Network (PCN)" or "Ward" instead of "Sub-ICB." The choice of boundary significantly dictates the "story" the map tells.
- **Time-Lagged Indices:** The Index of Multiple Deprivation (IMD) is only updated every few years (the last major set was 2019). Socioeconomic conditions in 2026 may have shifted significantly, particularly following the pandemic and inflation, making the "Need" score potentially outdated.

4. End-to-End Pipeline Technical Constraints

- **Memory vs. Granularity:** To avoid crashing local hardware with 6.8 GB files, we used "Chunking" and "Aggregation." The limitation here is the loss of granularity.

We can see *that* a CCG is an anomaly, but we cannot see *which specific doctor* or *which specific patient* is driving it without re-running the heavy, raw data.

- **Static vs. Dynamic:** This pipeline is currently a "snapshot." It reflects the state of the NHS in May 2026. Without a continuous data feed, these insights will lose their operational value within 3 to 6 months.

5. Clinical Recommendations

1. Workforce: Use the Upper Confidence Interval (192k referrals) for "Stress Testing" the service.
2. Finance: Investigate the top 1% of Isolation Forest flags to identify potential £100k+ savings in generic drug switching.
3. Equity: Direct mobile health units to the High-Need Low-Service areas identified in the spatial map to reduce health inequalities.